

BINAY - Lighting Up The Future



BINAY LED-based HIGH, MEDIUM AND LOW Intensity Aviation Obstruction Light Beacons

As per International Civil Aviation Organisation (ICAO) requirements Available in Low Intensity, Medium Intensity and High Intensity versions (as per International Civil Aviation Organisation guidelines), BINAY's patented LED Aviation Lights come with 5-year/3-year warranties.

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick Installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed Intensity to allow for natural LED intensity degradation over its operating lifetime



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The device and chipset issue

While the capacity to launch LTE in a wide gamut of spectrum bands has advantages, it brings the baggage of complications for chipmakers and device manufacturers. Which frequencies will be used by operators is a big question today as it will significantly impact costs, vendor margins, time to market and distribution, create technical challenges for global roaming and inhibit decision-making process within the operator community. It is important for chipset manufacturers and OEMs such as Qualcomm, Altair and Broadcom to understand the operator demand for LTE deployments by band so that they can then produce them on a large scale.

From the table given here, it is evident that a considerably larger number of devices, including dongles, routers, smartphones and tablets, are available on LTE-FDD, and more so in the 700MHz band, due to adaptability of the same in the US market. The LTE ecosystem is evolving at great speed with 347 LTE devices launched by 63 manufacturers available in the market as of April 2012.

Increased spectrum efficiency

Spectral efficiency is the information rate that can be transmitted over a given bandwidth in a specific communication system. It is a measure of how efficiently a limited frequency spectrum is utilised by the physical-layer protocol, and sometimes by the media access control (channel access protocol).

The LTE system can be scaled from 1.4 to 20 MHz; 1.4 MHz, 3 MHz, 5 MHz, 10 MHz, 15 MHz and 20 MHz wide cells are standardised. This means that apart from being able to operate in various frequency bands, it also promises scalable bandwidth. Networks can be launched with a small amount of spectrum, alongside existing services, and more spectrum can be added as the number of subscribers grows. It also enables operators to customise their network deployment as per their needs and available spectrum resources rather than being forced to make their spectrum fit a certain technology.

LTE can be deployed in clear spectrum with bandwidth as wide as 20 MHz of paired spectrum (20MHz uplink, 20MHz downlink). Paired spectrum is two equal parts of airwaves—one for sending and the other for receiving information—while unpaired spectrum is only a single part of airwaves meant to either receive or send information. Voice

Global LTE Device Availability

Spectrum	LTE-FDD	Spectrum	LTE-TDD
	Number of devices		Number of devices
700 MHz	142	2300 MHz	36
800 MHz	52	2600 MHz	41
1800 MHz	50		
2600 MHz	65		
800/1800/2600 MHz	43		

Source: Global Mobile Suppliers Association and CARE Research